🔄 Lehman Brothers: Resolution and Capital Recovery Analysis

PROVISIONAL: Recovery scores (R_p) are derived from publicly available Chapter 11 proceedings, the Valukas Examiner Report (2010), FCIC Final Report (2011), and Lehman Brothers Holdings Inc. Plan of Reorganization disclosure documents (2012). Omega weights reflect estimated economic significance of each phase to total stakeholder value at time of failure. Creditor recovery estimates use final distributions reported through the Lehman estate wind-down (as of 2016).

TVPCI (PRIMARY CHAIN)

49.9%

8-phase structural balance at failure

TVPCI-R (RESOLUTION RECOVERY)

25.1%

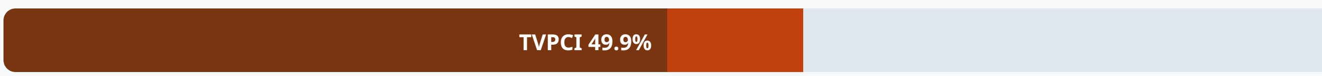
Weighted recovery score across phases

B_CHAIN (HOLON BALANCE)

37.0%

exp(-2 × |TVPCI - TVPCI-R| / max)

TVPCI VS TVPCI-R: PRIMARY CHAIN VS RECOVERY CHAIN BALANCE



Delta_chain = 49.9% - 25.1% = +24.8 percentage points. This is the largest primary-chain-to-recovery-chain gap in this project's dataset. Compare to SVB: Delta = +4.8 pp (FDIC systemic risk exception preserved depositor value). Lehman's absence of a government backstop and the complexity of a $613B Chapter 11 produced a recovery chain 24.8 points below the already-Failure primary chain. B_chain of 37.0% (below the 38.2% Breakdown threshold) indicates the resolution process itself was structurally in Breakdown relative to the primary chain health.

PHASE 0

Senior Unsecured Creditor Recovery

21%

ω = 0.30

Senior unsecured creditors recovered approximately 21 cents on the dollar through the Chapter 11 plan (2012-2016). $340B+ in claims settled over 8 years. No equivalent of the FDIC systemic risk exception that made SVB depositors whole.

PHASE 1

Equity / Capital Recovery

0%

ω = 0.10

Equity completely wiped out. Shareholders received nothing. Subordinated debt holders received minimal recovery. Standard outcome in Chapter 11 reorganization at this scale.

PHASE 2

Asset Recovery (CRE and Mortgage)

40%

ω = 0.25

Commercial real estate and mortgage portfolios sold at deep discounts during and after the crisis. CRE values declined 30-50% from peak. Estimated 40% recovery on illiquid asset base: better than feared at time of failure as markets partially recovered over the multi-year wind-down.

PHASE 3

Leverage Risk Resolution

15%

ω = 0.15

Repo counterparties suffered losses as collateral values declined. Tri-party repo clearing disputes took years to resolve. Estimated 15% recovery on the structural leverage risk phase: reflecting both actual losses and the multi-year legal cost of unwinding $200B+ in repo agreements.

PHASE 4

Liquidity Recovery

55%

ω = 0.05

PHASE 5

Franchise and Business Recovery

25%

PHASE 6

Market Confidence

5%

ω = 0.05

PHASE 7

Regulatory Learning

45%

ω = 0.05

Barclays acquired the US broker-dealer (Lehman Brothers Inc.) for approximately $1.75B, preserving some client account continuity. SIPC provided limited coverage for individual investors. No FDIC bridge bank equivalent: liquidity function was partially restored by asset sale rather than government backstop.

ω = 0.05

Barclays acquired investment banking and broker-dealer operations; Nomura acquired Asia-Pacific and EMEA operations. Franchise value was substantially destroyed. Estimated 25% recovery on the institutional franchise: better than zero but representing a fraction of the $600B+ pre-failure balance sheet scale.

The Lehman failure triggered the acute phase of the global financial crisis. AIG required an $85B government bailout the following day. Money market funds "broke the buck." Global credit markets froze for weeks. Institutional confidence in unguaranteed wholesale funding was permanently impaired. Market confidence phase: effectively unrecoverable.

Dodd-Frank Act (2010) enacted Title I (SIFI designation), Title II (Orderly Liquidation Authority), Title VII (derivatives reform). Resolution planning requirements, enhanced prudential standards, and FSOC created. Substantial reform accomplished; full implementation ongoing as of 2026. Scored 45% reflecting incomplete Dodd-Frank implementation and partial rollback of some provisions.

Value Loss Table: Phase-by-Phase Economic Impact

PHASE	PRIMARY VALUE LOST	ESTIMATED SCALE	CURRENT DISPOSITION	RECOVERY POTENTIAL	R_P
Ph 0: Funding Base	Senior unsecured creditor principal	$340B+ in unsecured claims	~21 cents/dollar recovered through Chapter 11 plan (2012-2016)	Achieved: 8-year wind-down	**21%**
Ph 1: Equity	Shareholder and subordinated debt capital	~$28B equity wiped out	Zero recovery for equity holders	None: Chapter 11 requires absolute priority	**0%**
Ph 2: Assets	CRE and mortgage portfolio discount at distressed sale	$57B peak CRE/RMBS; 30-50% decline from peak	Sold progressively by estate; partial recovery as markets improved 2010-2014	Partial: time and market recovery aided estate	**40%**
Ph 3: Leverage Risk	Repo counterparty losses and collateral disputes	$200B+ repo contracts; years of dispute resolution	Tri-party repo disputes resolved; significant legal and haircut losses absorbed by counterparties	Low: structural losses crystallized at failure	**15%**
Ph 4: Liquidity	Client account disruption; prime	Hundreds of billions in prime brokerage	Barclays broker-dealer acquisition	Partial: no FDIC bridge bank	**55%**

PHASE	PRIMARY VALUE LOST	ESTIMATED SCALE	CURRENT DISPOSITION	RECOVERY POTENTIAL	R_P
	brokerage freeze	assets temporarily frozen	restored partial access; SIPC provided individual investor coverage	equivalent available	
Ph 5: Franchise	Lehman brand, advisory relationships, technology platform	Substantial intangible and franchise value destroyed	Barclays (US) and Nomura (EMEA/Asia) acquired operations at distressed prices	Low: institutional franchise does not survive Chapter 11 intact	**25%**
Ph 6: Confidence	Global credit market stability; systemic contagion	Estimated $10-20 trillion in global wealth destruction; $700B+ in government interventions globally	Global financial crisis; Fed/Treasury emergency programs; G20 Basel III reforms	None for Lehman specifically; systemic stability rebuilt over 2009-2013	**5%**
Ph 7: Regulation	Lost opportunity for pre-failure supervisory reform	20 months of systemic exposure during Breakdown period unaddressed	Dodd-Frank enacted; FSOC, OLA, SIFI designation, Basel III all implemented	Partial: significant reform accomplished; full Dodd-Frank implementation ongoing	**45%**

Counterfactual: What Early Intervention Would Have Recovered

If the N-D-C structural signal had been acted upon in Q1 2007 (the quarter Phase 3 first crossed into Breakdown at 29.3%), the following resolution path was available:

- Supervisory requirement to reduce gross leverage from 30x+ to 20x over 4 quarters: Phase 3 balance recovery to approximately 50-60%. Asset sales at 2007 valuations, before the acute crisis phase, would have recovered significantly more than distressed 2008-2009 prices.
- Mandatory minimum term funding ratio of 30% enforced from Q1 2007: Phase 0 recovery to approximately 65-70%. Overnight repo dependency reduced from 80% to below 60%, making a terminal counterparty withdrawal run structurally impossible at the same velocity.
- CRE and mortgage concentration cap at 15%: Phase 2 forced reduction from $57B to approximately $40B in correlated illiquid positions, at 2007 prices with active secondary markets

rather than 2008-2009 fire-sale conditions. Estimated additional recovery of $8-12B on asset portfolio.

Estimated counterfactual outcome: No Chapter 11 filing. Creditors recover approximately 55-65 cents on the dollar in an orderly wind-down rather than 21 cents over 8 years. AIG bailout potentially avoidable (Lehman's failure triggered the AIG margin call cascade). Global financial crisis significantly less severe: conservatively $500B-2T in global economic cost avoided by preventing the acute confidence collapse of September-October 2008. Total economic value of early intervention: the FCIC estimated the US economic cost of the crisis alone at $22 trillion in lost household wealth; even a 10% reduction in that cost from earlier supervisory action would represent $2.2 trillion in preserved value.

Resolution Comparison: Lehman vs SVB

DIMENSION	LEHMAN (2008)	SVB (2023)
TVPCI-R (recovery score)	**25.1%**	**56.5%**
B_chain (holon balance)	**37.0%: Breakdown**	**85.5%: Good**
Depositor/creditor outcome	Senior unsecured: ~21 cents/dollar over 8 years	All depositors made whole within 48 hours
Government resolution tool	Chapter 11 (no FDIC/systemic risk tool available)	FDIC systemic risk exception
Systemic contagion	Global financial crisis; $700B+ in government interventions worldwide	Signature Bank contagion; Fed BTFP backstop; contained within weeks
Resolution timeline	8+ years of wind-down proceedings	First Citizens acquisition completed in weeks
Regulatory reform outcome	Dodd-Frank; Basel III; SIFI designation; OLA	Enhanced LCR extension; AOCI proposals; partial implementation

The B_chain score encapsulates the resolution quality difference. SVB at 85.5% reflects a well-structured resolution that preserved most stakeholder value (depositors) despite institutional failure. Lehman at 37.0% (below Breakdown threshold) reflects a resolution that could not preserve stakeholder value: not due to regulatory failure alone, but because no statutory resolution authority existed for non-bank broker-dealers at the time.

Priority Resolution Interventions: Applicable to Future Crises

INTERVENTION	PHASE	DESCRIPTION	EXPECTED R_P IMPACT
Orderly Liquidation Authority (Dodd-Frank Title II)	Phase 0 and 4	Provides FDIC-equivalent resolution authority for systemically important non-bank institutions. Enacted post-Lehman. Would have enabled bridge institution with government backstop instead of Chapter 11. Now available but not yet tested at Lehman scale.	Phase 0 R_p: 21% to estimated 50-60% (creditor recovery improves with orderly liquidation). Phase 4 R_p: 55% to 75-80% (government bridge institution provides liquidity continuity).
Pre-positioned resolution plan with legal entity separation	System-wide	Dodd-Frank §165(d) living wills require systemically important firms to pre-position for orderly wind-down. Resolution plans that separate broker-dealer client assets from holding company liabilities prevent client asset freezes. Key lesson from Lehman: prime brokerage client assets should have been segregated in advance.	System TVPCI-R improves from 25.1% to estimated 40-50% (orderly transfer preserves business continuity and prevents contagion cascade).
Mandatory leverage reduction trigger at Phase 3 Breakdown detection	Phase 3	Any broker-dealer whose N-D-C Phase 3 balance falls below 38.2% for two consecutive quarters triggers a mandatory supervisory deleveraging requirement: reduce gross leverage to below 25x within 90 days or face suspension of dividend and buyback authority. This closes the 20-month gap between structural detection (Q1 2007) and terminal failure (September 2008).	Phase 3 R_p: 15% to estimated 55-70% (structural failure prevented by early deleveraging rather than terminal collapse). System TVPCI-R: 25.1% to estimated 45-55%.